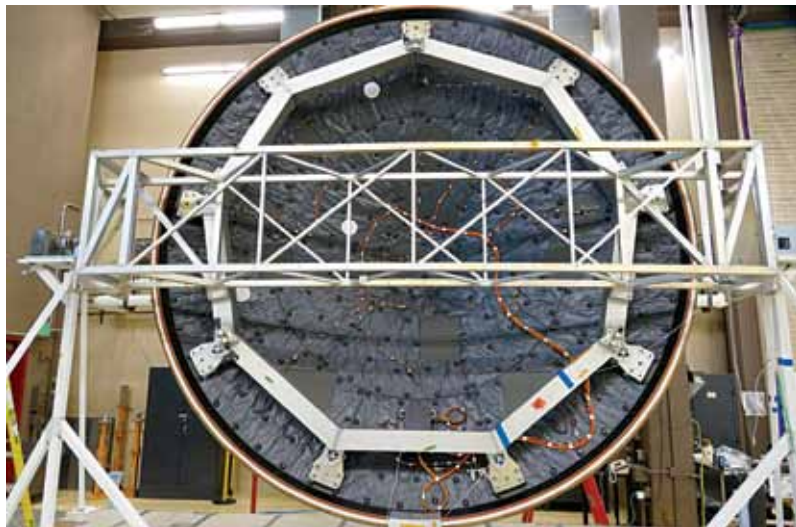


Sensors aboard Mars rover collect atmospheric data



MEDLI2 sensors are installed on the Mars 2020 heat shield and back shell prior that will protect NASA's Perseverance rover on its journey to the surface of Mars (Credit: www.jpl.nasa.gov)

Nearly six-and-a-half months since launch, NASA's Perseverance rover has begun its robotic exploration of the Red Planet. And to assist in a successful entry, descent, and landing have been the rover's Mars Entry, Descent, and Landing Instrumentation 2 (MEDLI2) aeroshell, which can collect critical data about the harsh entry environment throughout the planet's atmosphere.

The MEDLI2 includes three types of sensors—thermocouples, heat flux sensors, and pressure transducers—for measuring extreme heat and pressure during entry. It also contains electronics and hardware for recording the thermal and pressure loads experienced during entry and through parachute deployment. It is hoped that the MEDLI2 data will help improve the designs of entry systems for future robotic and crewed Mars missions.

Photonic D-A converter developed for next-gen hardware

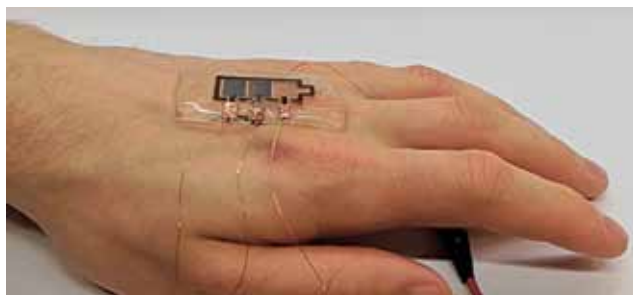
Current optical networks as well as many sensors require a digital-to-analogue conversion for data transmission. Now researchers at the George Washington University and the University of California, Los Angeles, have developed and demonstrated a photonic digital-to-analogue converter that does not require the signal to be converted in the electrical domain.

The converter meets high data-processing capabilities while acting on optical data, interfacing to digital systems and performing in a compact footprint, with both short signal delay and low power consumption. Such novel converters can advance next-generation data processing hardware with high relevance for data centres, 6G networks, artificial intelligence, and more.



Researchers create Novel Photonic chip (Credit: <https://mediarelations.gwu.edu>)

Electronic displays that are biodegradable



Thanks to its flexibility and adhesion, the biodegradable display can be worn directly on the hand (Credit: www.kit.edu/kit)

To save resources and minimise waste caused by the increasing use of electronic devices, an eco-friendly and more sustainable life cycle is needed. Now, scientists of the Karlsruhe Institute of Technology have produced displays that are largely based on natural materials. After use, these displays can be safely composted.

The functioning of the display is based on the electrochromic effect of the initial organic material. When voltage is applied, light absorption is modified, and the material changes its colour. These displays have a low energy consumption compared to LED, LCD, and e-paper. Another advantage is that these displays can be inexpensively produced by inkjet printing.